

GRAPH NEURAL NETWORKS

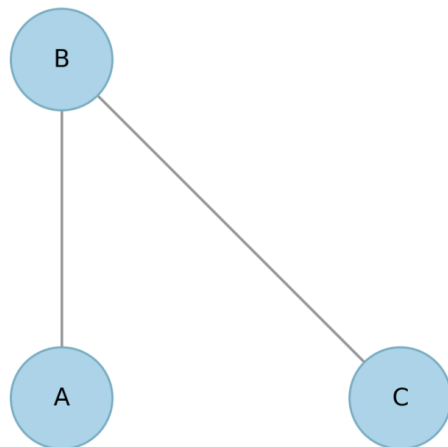
AI506 — Advanced Machine Learning

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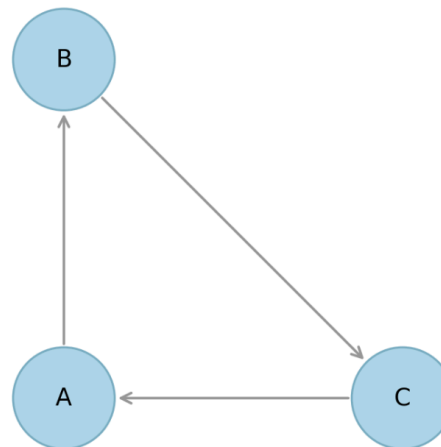
May 2026

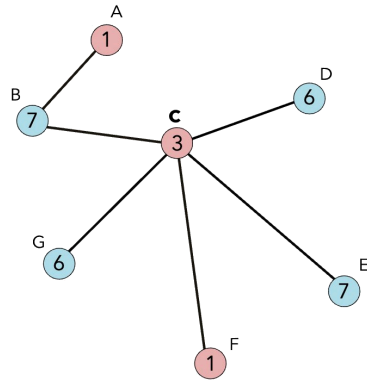
- Nodes: V
- Node features: $V \rightarrow \mathbb{R}^d$
- Edges: $E \subseteq V \times V$

Undirected



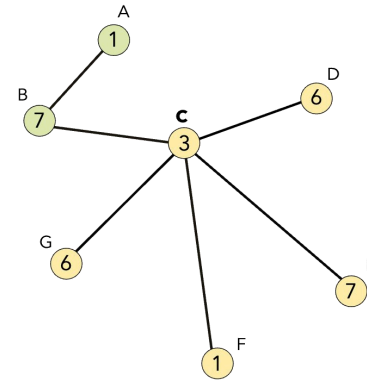
Directed





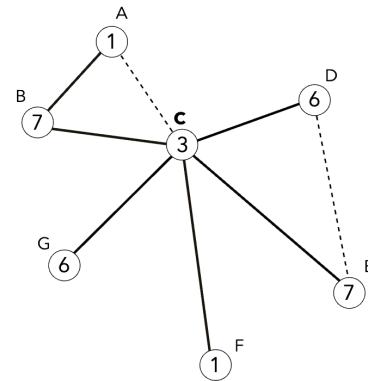
Node Classification

Toxic

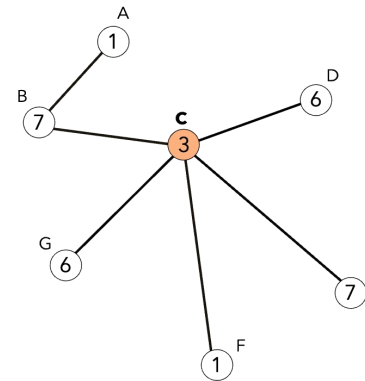


Node Clustering

Graph Classification



Link Prediction



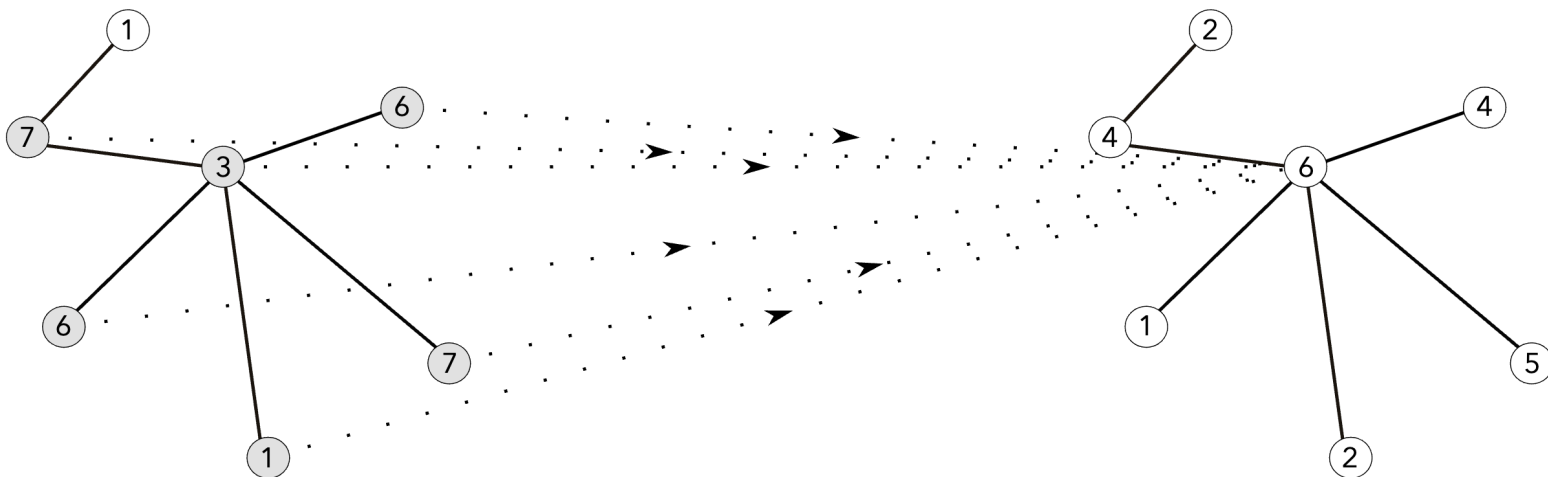
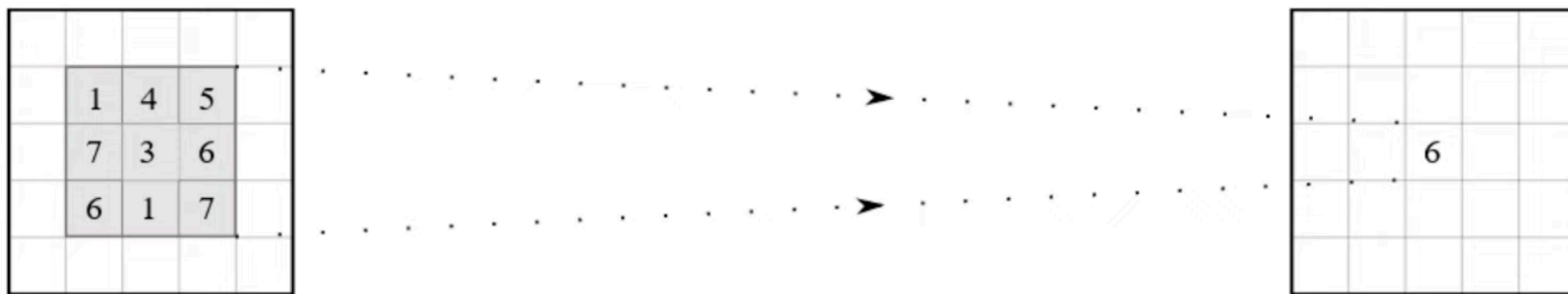
Influence Maximization

For node classification:

Let $X = H^0$

$$H^{(l+1)} = \sigma(\hat{A}H^{(l)}W^{(l)} + b^{(l)})$$

$$Z = \text{softmax}(\hat{A} \text{ReLU}(z^{(l-1)})W^{(l)})$$



$$h_i^{(l+1)} = \sigma\left(\text{UPD}\left(h_i^{(l)}, m_i^{(l)}\right)\right)$$

$$\text{UPD}\left(h_i^{(l)}, m_i^{(l)}\right) = W_h \cdot h_i^{(l)} + W_{\mathcal{N}} m_i^{(l)} + b^{(l)}$$

where

$$m_i^{(l)} = \text{AGG}_{j \in \mathcal{N}(i)} g\left(h_i^{(l)}, h_j^{(l)}\right)$$

g = some transformation of h_i and h_j

$$\text{AGG}_{\text{sum}} = \sum_{j \in \mathcal{N}(i)} h_j^{(l)}$$

$$\text{AGG}_{\text{max}} = \max_{j \in \mathcal{N}(i)} \left(h_j^{(l)} \right)$$

$$\text{AGG}_{\text{avg}} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} h_j^{(l)}$$

$$\text{AGG}_{\text{norm}} = \sum_{j \in \mathcal{N}(i)} \frac{h_j^{(l)}}{\sqrt{(\|\mathcal{N}(i)\| \cdot \|\mathcal{N}(j)\|)^2}}$$

- Edge attributes (weight)

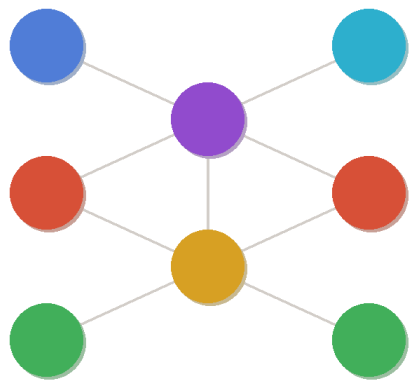
$$g\left(h_i^{(l)}, h_j^{(l)}\right) = a_{ij} \cdot h_j^{(l)}$$

$$\text{where } a_{ij} = \text{softmax}_{\mathcal{N}(i)}\left(a^T [Wh_i \parallel Wh_j]\right)$$

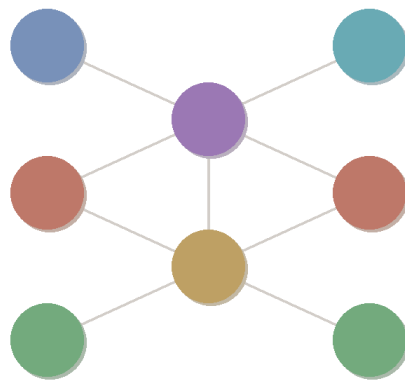
Where a is a learnable vector.

$$\text{AGG}_{\text{sum}} = \sum_{j \in \mathcal{N}(i)} a_{ij} h_j^{(l)}$$

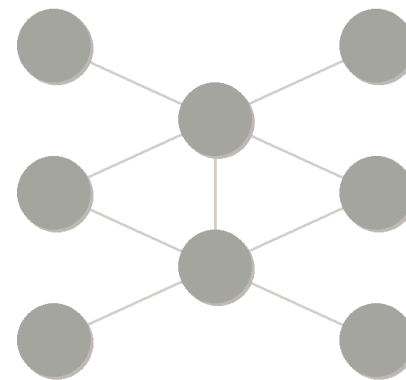
$$\text{AGG}_{\text{avg}} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} h_j^{(l+1)} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} \left(\frac{1}{\|\mathcal{N}(j)\|} \sum_{k \in \mathcal{N}(j)} h_k^{(l)} \right)$$



Layer 0



Layer 1



Layer 2

Concatenation:

$$z_i = \text{MLP} \left(h_i^{(1)} \# h_i^{(2)} \# \dots \# h_i^{(l)} \right)$$

Max pooling:

$$z_i = \text{MAX} \left(h_i^{(1)}, h_i^{(2)}, \dots, h_i^{(l)} \right)$$

LSTM:

$$z_i = \text{LSTM} \left(h_i^{(1)}, h_i^{(2)}, \dots, h_i^{(l)} \right)$$